SUPPORTING ADULT GUIDANCE

Simplifying algebraic fractions

Algebraic Expressions and Identities · Lesson 4. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson applies the rules of number fractions to algebra. Learners first see $(3 \times 5)/3 = 5$ but $(3 + 5)/3 \neq 1 + 5$, then practise factorising the expressions they will need. In the main activity, numerators and denominators appear as factor tiles: learners tap matching tiles to cancel them, then type what is left. They then test tempting shortcuts against a table of values, where a false cancellation produces a counterexample. In Creativity they disguise a target inside a fraction with a quadratic.

Driving Question: *When can I cancel, and when can't I?*

The mathematics, in plain terms

For polynomials A, B, C with $C \neq 0$, $AC/(BC) = A/B$, because $C/C = 1$. Only common factors of the whole numerator and the whole denominator may be cancelled; a term that is added is not a factor. The simplified fraction agrees with the original everywhere except where the cancelled factor is zero, which must be excluded from the domain.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Factorise, then cancel

Simplify $(2x + 6)/(x^2 - 9)$.

Factor tiles

2	~~$(x + 3)$~~
~~$(x + 3)$~~	$(x - 3)$

1. Factorise the numerator.	$2x + 6 = 2(x + 3)$
2. Factorise the denominator: a difference of two squares.	$x^2 - 9 = (x + 3)(x - 3)$
3. Cancel the common factor $(x + 3)$.	$(x + 3)/(x + 3) = 1$
4. What is left.	$2/(x - 3)$

$(2x + 6)/(x^2 - 9) = 2/(x - 3)$, for $x \neq \pm 3$.

Example 2 — Two quadratics

Simplify $(x^2 - 4)/(x^2 + 4x + 4)$.

Factor tiles

~~$(x + 2)$~~	$(x - 2)$
~~$(x + 2)$~~	$(x + 2)$

1. Factorise both.	$(x + 2)(x - 2)$ over $(x + 2)(x + 2)$
2. One factor of $(x + 2)$ cancels with one.	the other $(x + 2)$ stays
3. What is left.	$(x - 2)/(x + 2)$
4. Check with $x = 1$: $(1 - 4)/(1 + 4 + 4) = -3/9$, and $(1 - 2)/(1 + 2) = -1/3$.	✓

$(x^2 - 4)/(x^2 + 4x + 4) = (x - 2)/(x + 2)$, for $x \neq -2$.

Numbers first, then algebra

Every rule here is one learners already trust for numbers. The tiles make 'whole factors only' physical — a term inside a bracket simply cannot be tapped — and the table of values turns Lesson 1's counterexample idea into a test for any shortcut. Excluded values are noted lightly; they return in Lesson 8 with domains.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The number bridge: (3 × 5)/3 cancels; (3 + 5)/3 does not.	~5 min
Movement	Factorise to find the factors.	~7 min
Reflection	The cancel checker: factor tiles, then tempting shortcuts tested with values. Stretch: why factors cancel and terms do not.	~19 min
Creativity	Hide it in a fraction: disguise a target with a quadratic.	~9 min
Rest	Cancelling as noticing that something over itself is 1.	~2 min
Nutrition	Intellectual: careful trust in a shortcut.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Cancelling terms: $(x + 3)/3 = x + 1$.
Ask: “Try $x = 3$. Are the two sides equal?”

Cancelling the x in $(x + 5)/(x + 7)$.
Ask: “Is x multiplying the whole numerator?”

Cancelling before factorising.
Ask: “Can you write the numerator as a product first?”

Cancelling both factors of $(x + 1)^2$ with one in the numerator.
Ask: “How many factors of $(x + 1)$ are on each side of the line?”

Forgetting excluded values.
Ask: “Which value of x makes the original denominator zero?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: simplifying number fractions, and factorising. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner simplifies $(x^2 + 4)/(x + 2)$ to $x + 2$. The aim is a question that lets them find the problem themselves.

- “Can you factorise $x^2 + 4$?”
- “Try $x = 1$ in both. What do you get?”

- “Which expression would give $(x + 2)$ as a factor?”
- “So does this fraction simplify?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is $6x/9$ accepted for $6x^2/9x$?
It is equal, but not fully simplified; the lesson asks the learner to cancel the 3 as well, giving $2x/3$.

Why mention excluded values?
The simplified form is defined at values where the original is not. This prepares for domains in Lesson 8.

Is the proof required?
It is a stretch. The secure core is: factorise, then cancel whole factors.

What comes next?
Lesson 5: adding and subtracting algebraic fractions.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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